

ECEN 215, Spring 2022

Electrical Circuit Theory

Lecture 6: Thevenin, Norton, Power Transfer

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Assignments



- Assignment #2 due date is Today
- Assignment #3 due date is Next Wednesday

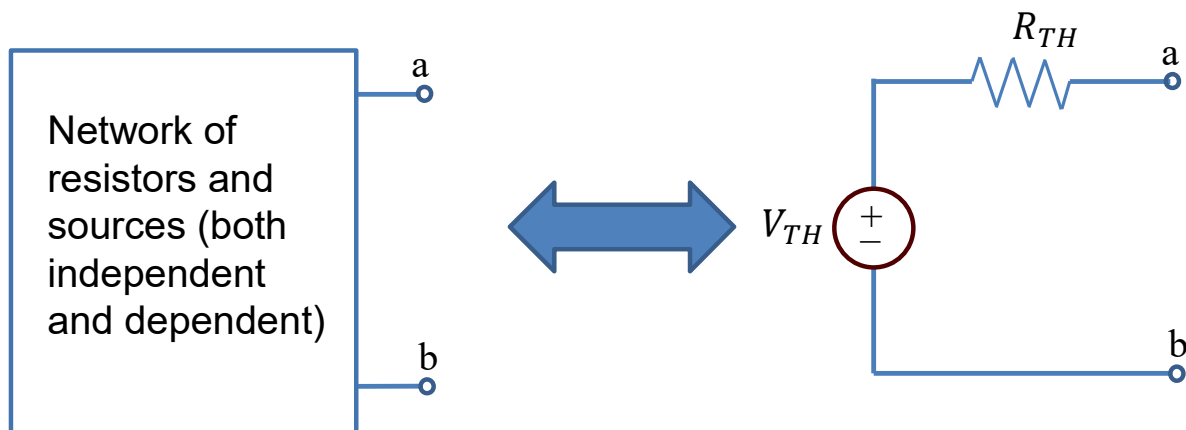
OUTLINE



- Mesh Circuit Analysis
 - Superposition
 - Source transformation
-

Thevenin Equivalent Circuits

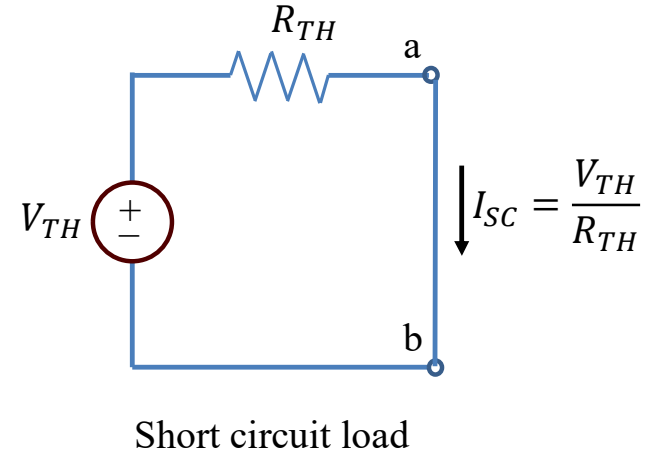
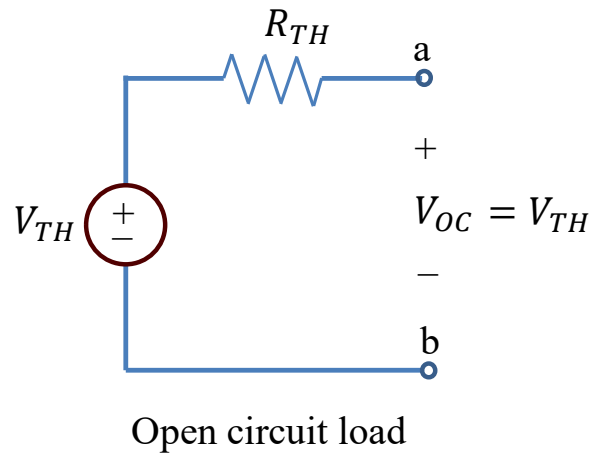
- A complicated circuit of sources and resistive elements can be reduced to a single voltage source in series with a resistor to study how the circuit will behave with respect to a load. This is known as a **Thevenin equivalent circuit**.



- Q:** How do we find the values of V_{TH} and R_{TH} so the equivalence holds?
- A:** Since there are only two unknowns, any two measurements that we can make on our circuit, should provide enough information to infer the Thevenin Equivalent circuit.

Thevenin Equivalent Circuits

- In order to find the appropriate values of V_{TH} and R_{TH} , consider loading the circuit with either an open circuit or a short circuit.



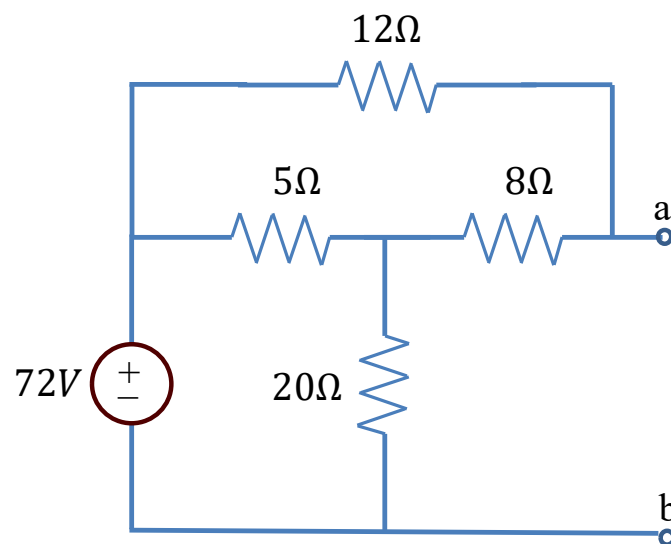
- From the two relationships above, the Thevenin equivalent circuit can be found from the open circuit voltage, V_{OC} , and the short circuit current, I_{SC} .

$$V_{TH} = V_{OC} \text{ and } R_{TH} = \frac{V_{OC}}{I_{SC}}$$

Example 1



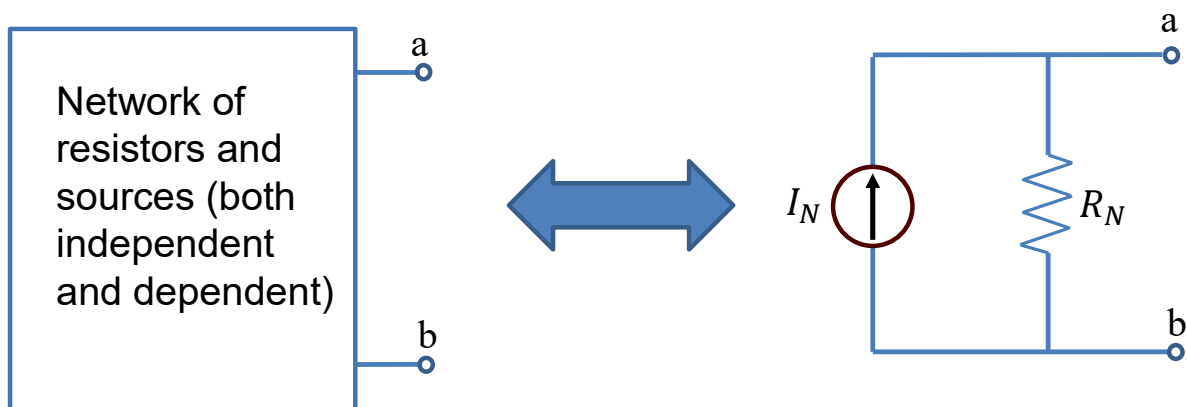
- Find the Thevenin equivalent with respect to the terminals a-b.



Norton Equivalent Circuits

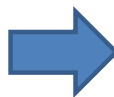


- Similar to a Thevenin equivalent circuit, we can also construct a **Norton equivalent circuit** which consists of a current source in parallel with a resistor.



$$V_{OC} = I_N R_N$$

$$I_{SC} = I_N$$



$$I_N = I_{SC}$$

$$R_N = \frac{V_{OC}}{I_{SC}}$$



same as R_{TH}



Thevenin and Norton Equivalent Circuits

There are a number of techniques we can use to find a Thevenin/Norton equivalent circuit.

Method 1: Use source transformations and serial/parallel combinations to reduce the circuit to its Thevenin or Norton equivalent.

Method 2: Find the open circuit voltage and the short circuit current. Then use:

$$V_{TH} = V_{OC}, I_N = I_{SC}, R_{TH} = R_N = \frac{V_{OC}}{I_{SC}}$$

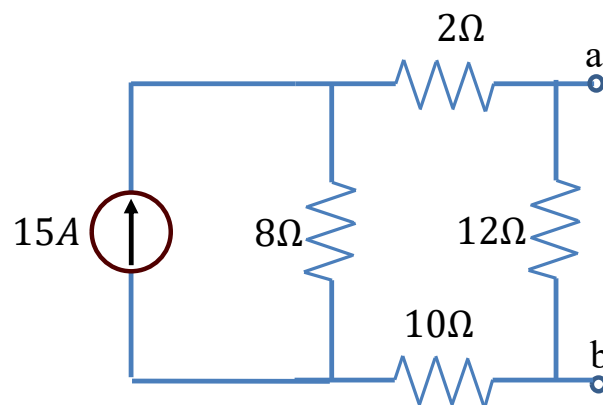
Method 3: For circuits with no dependent sources, R_{TH} can be found by disabling all independent sources ($V = 0 \rightarrow$ short, $I = 0 \rightarrow$ open) and computing the equivalent resistance as seen across the load terminals.

Note: Method 3 does not actually find the entire equivalent circuit. It only provides a shortcut for finding R_{TH} so that you only need to find V_{OC} **or** I_{SC} (not both) when using Method 2.

Example 2



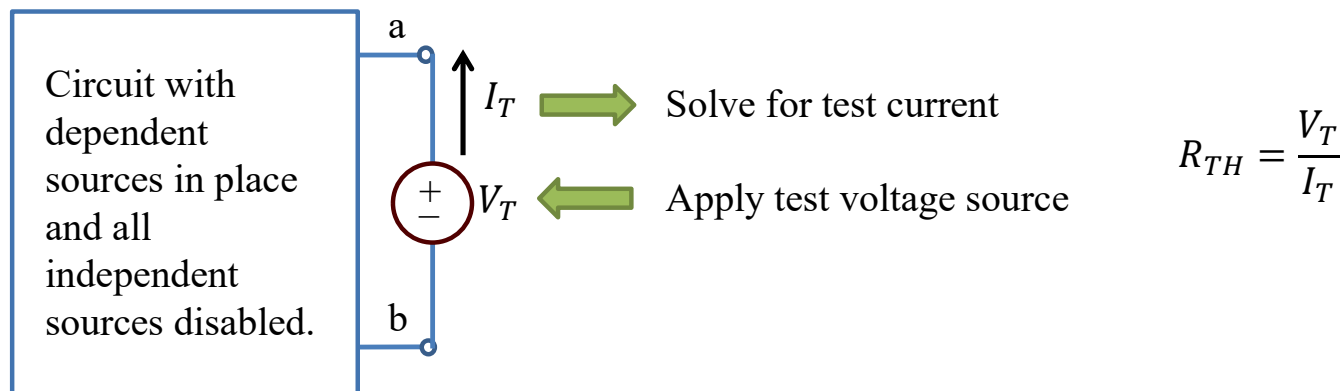
- Find the Thevenin and Norton equivalent circuits with respect to the terminals a-b.





Thevenin and Norton Equivalent Circuits

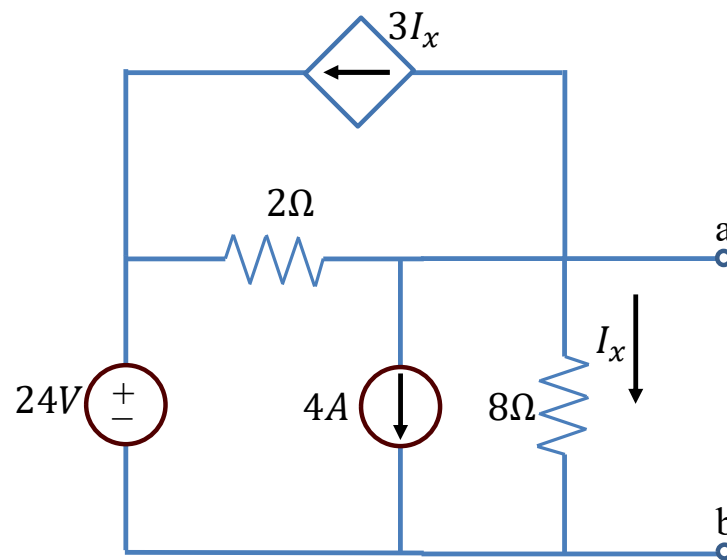
- For circuits with dependent sources, a modified version of Method 3 can be used.
- Method 3 (modified):** Disable all independent sources. Apply a test source to the load (either V_T or I_T) and then find I_T or V_T so that $R_{TH} = \frac{V_T}{I_T}$.



Example 3

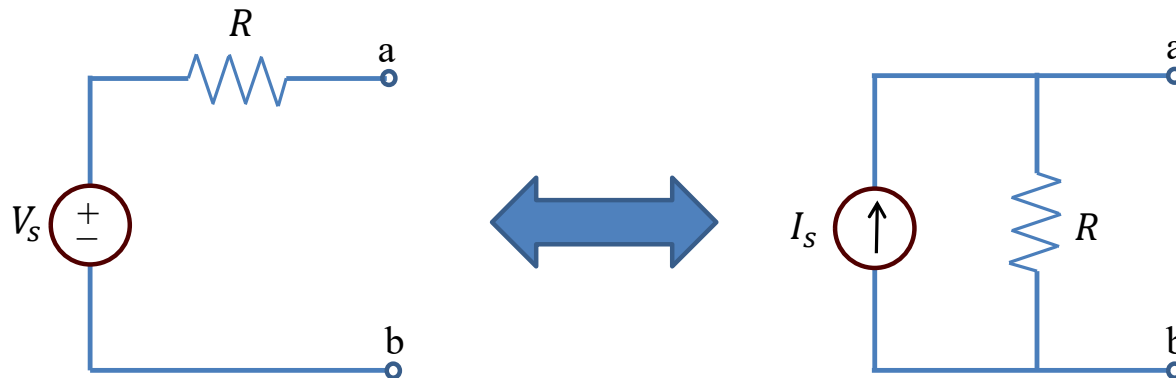


- Find the Thevenin equivalent with respect to the terminals a-b.



Source Transformations

- Often it is useful to transform a Thevenin Equivalent (series combination of a voltage source and a resistor) to a Norton Equivalent (parallel combination of current source and a resistor) or vice versa as shown below.

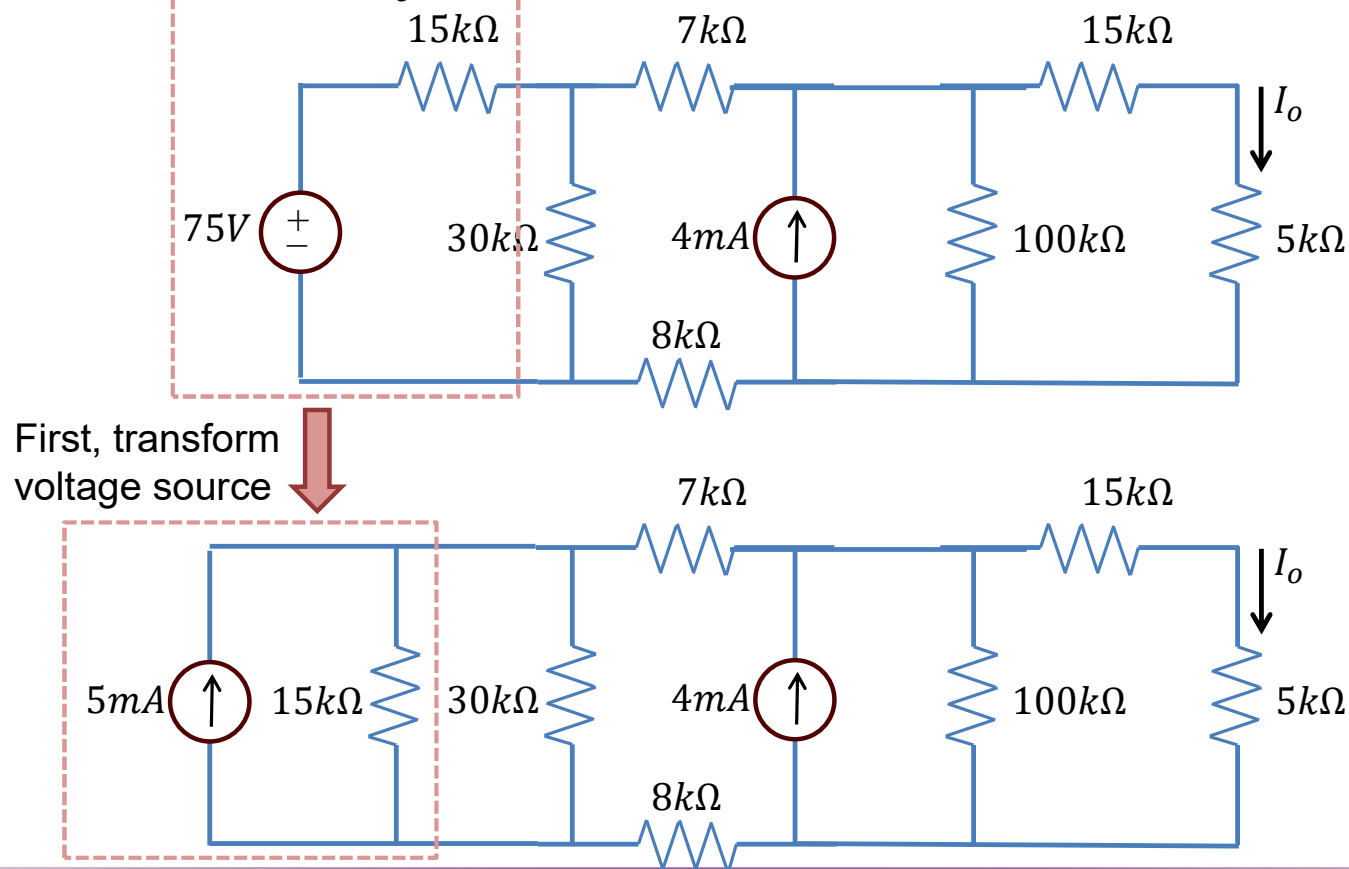


- These two are equivalent if $V_s = I_s \cdot R$

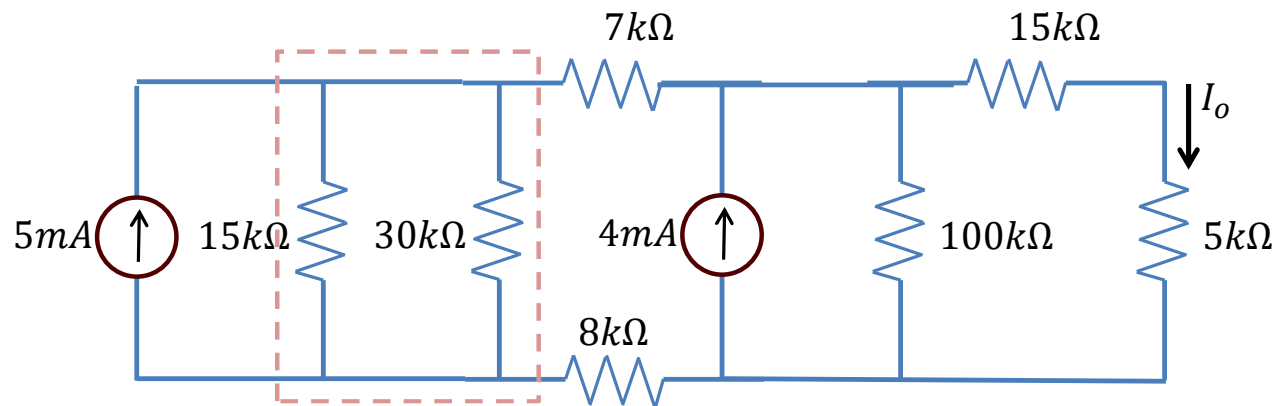
Example 4



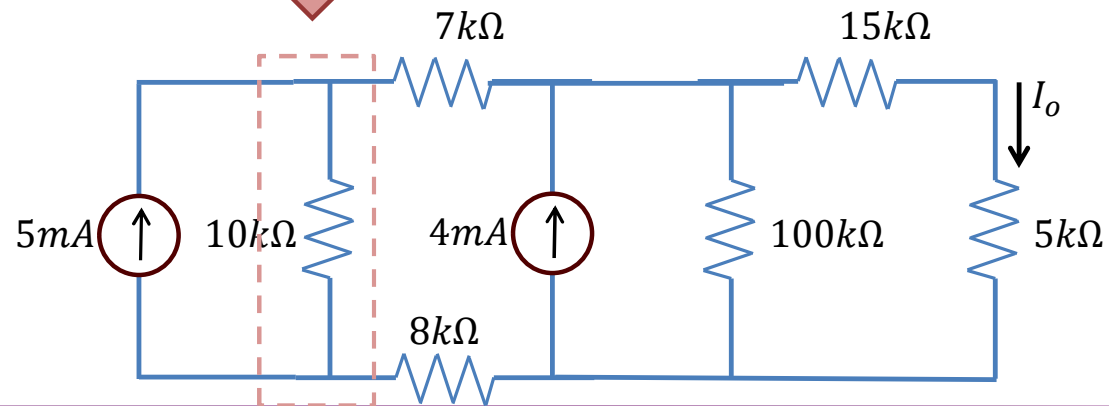
- Example:** Find the current, I_o , in the $5k\Omega$ resistor.



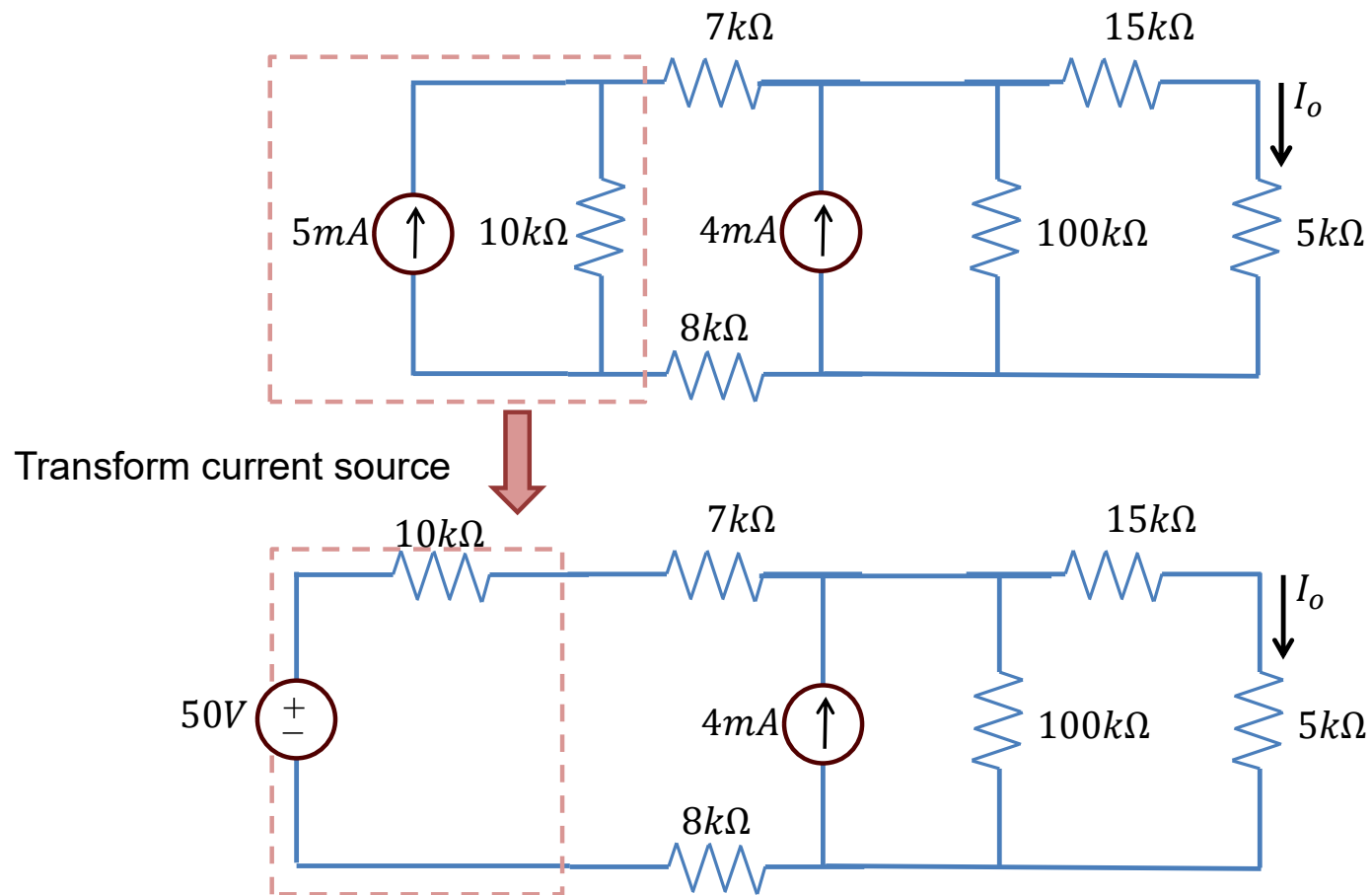
Example 4, cont.



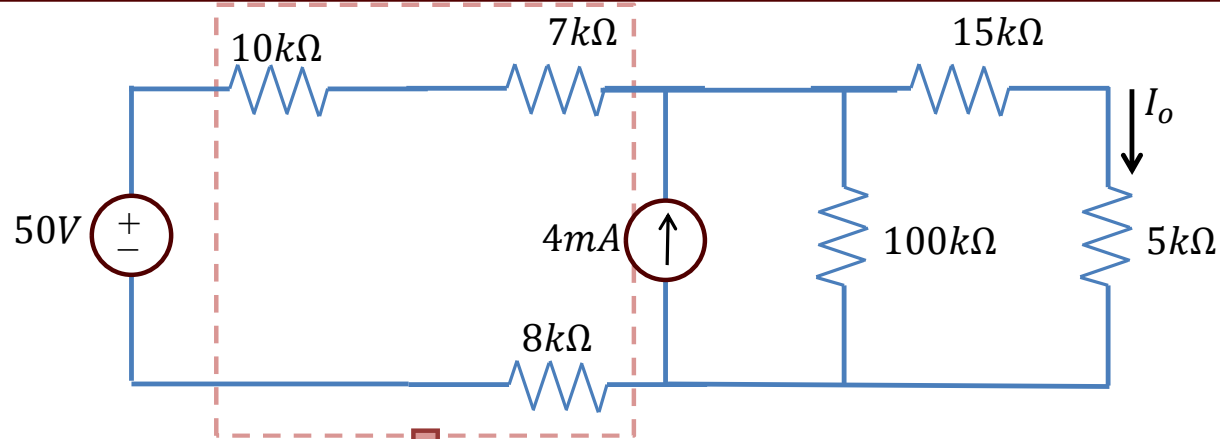
Combine parallel resistors



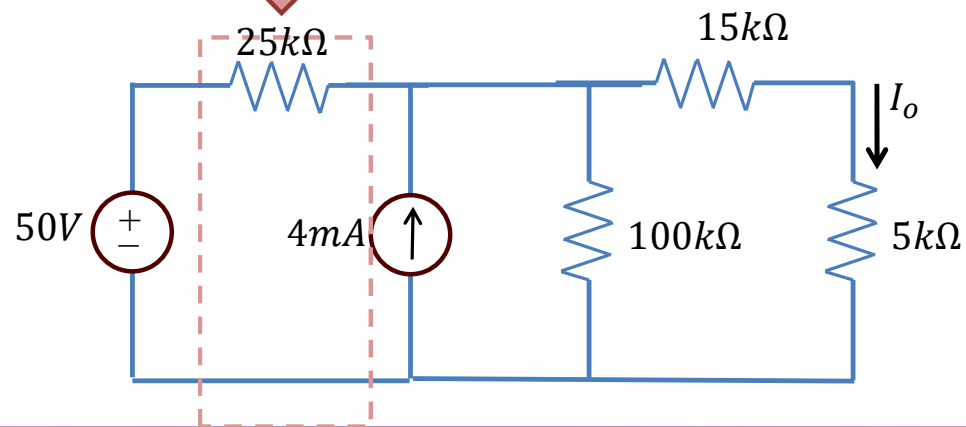
Example 4, cont.



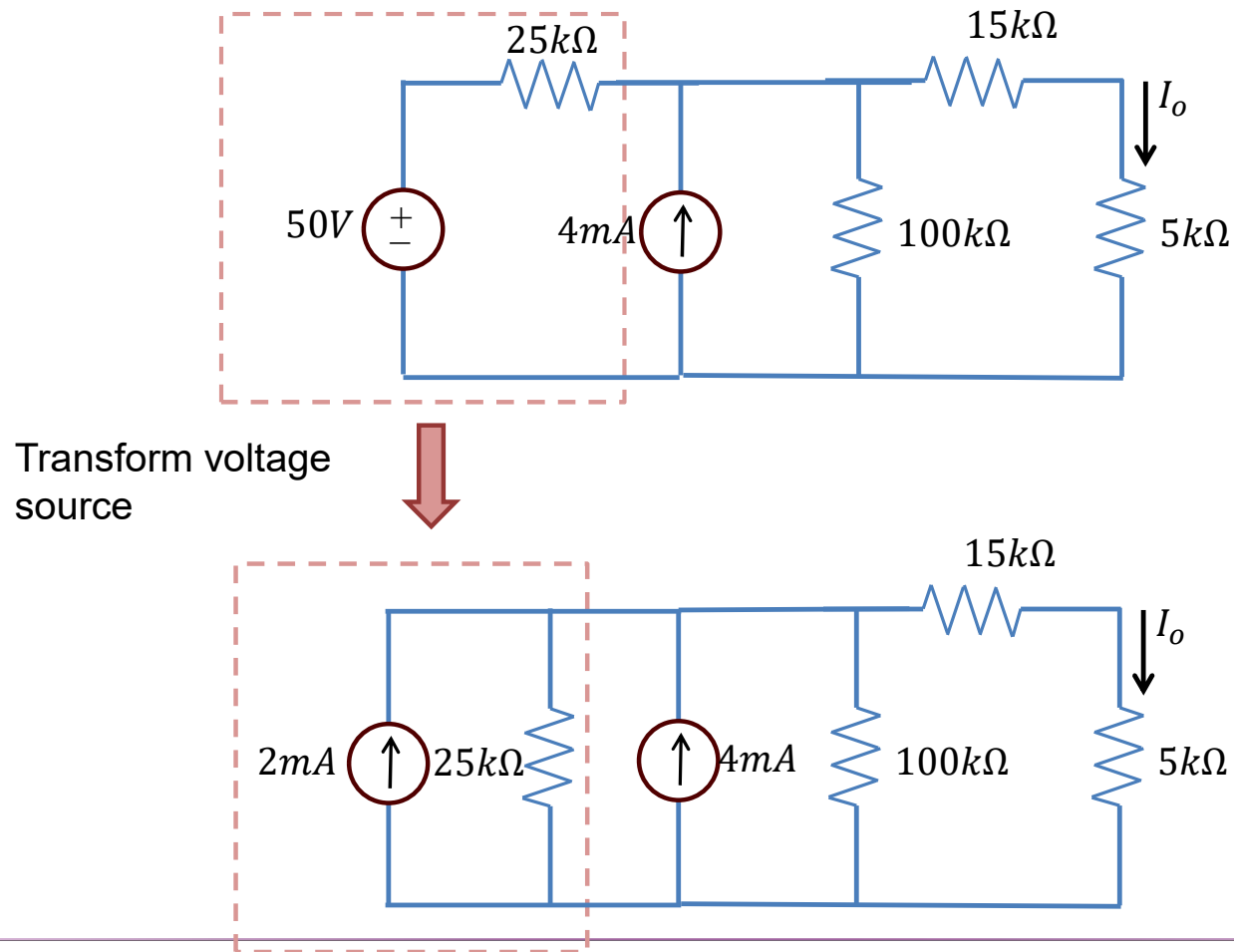
Example 4, cont.



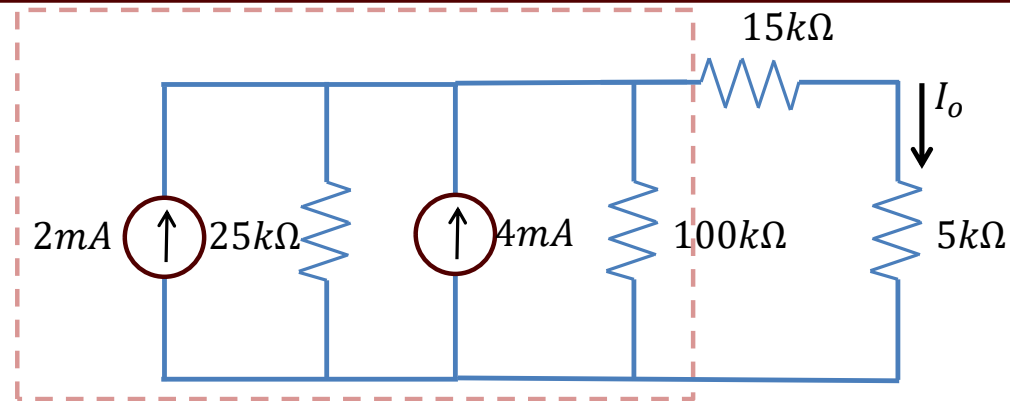
Combine series resistors



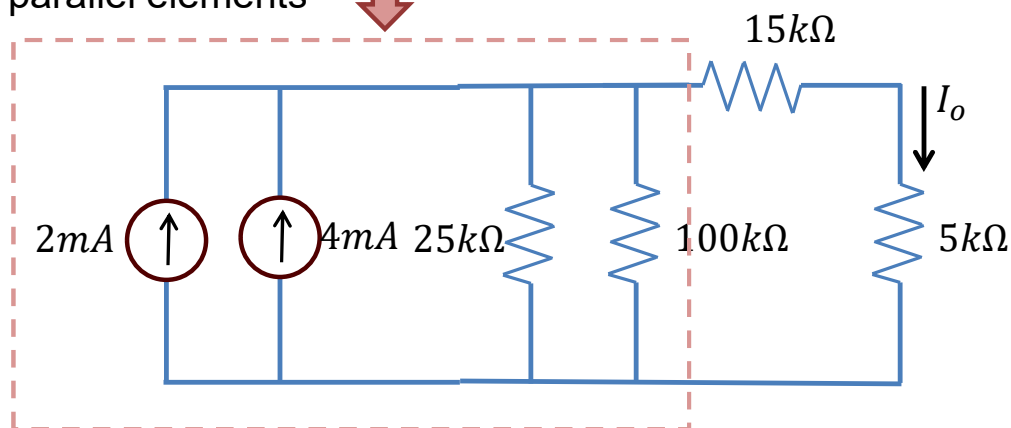
Example 4, cont.



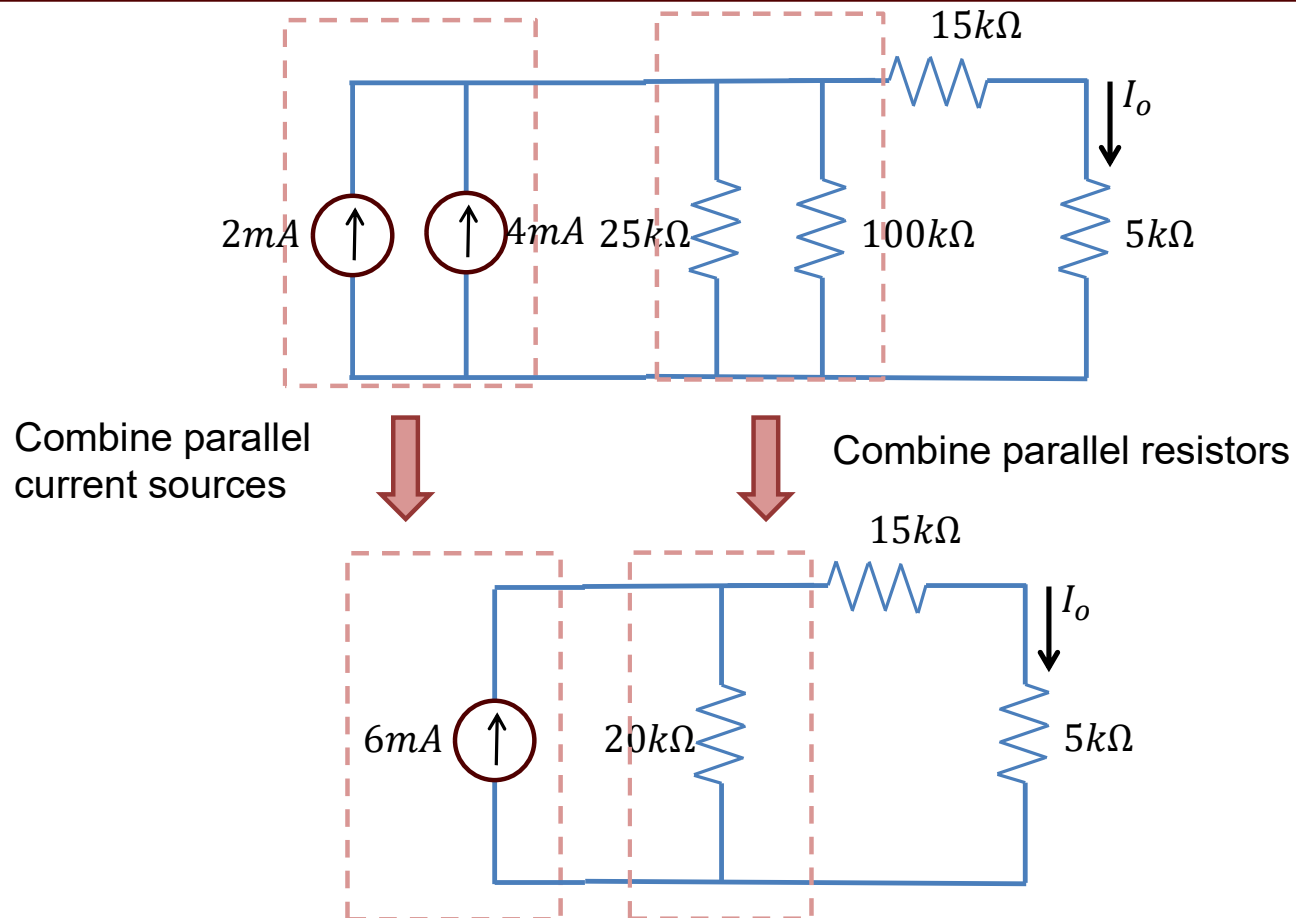
Example 4, cont.



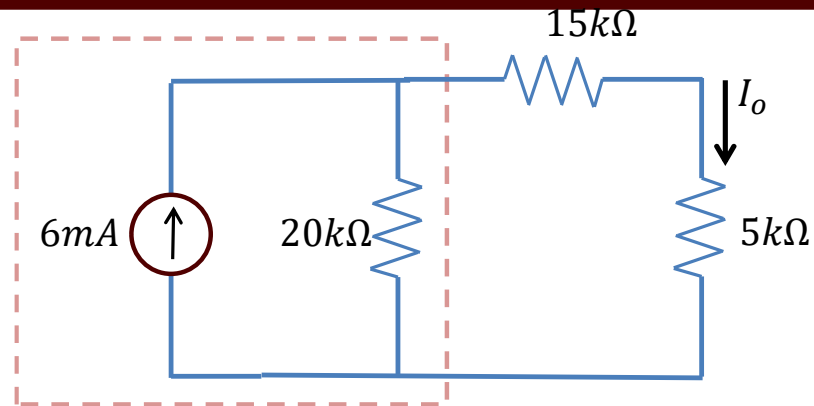
Rearrange
parallel elements



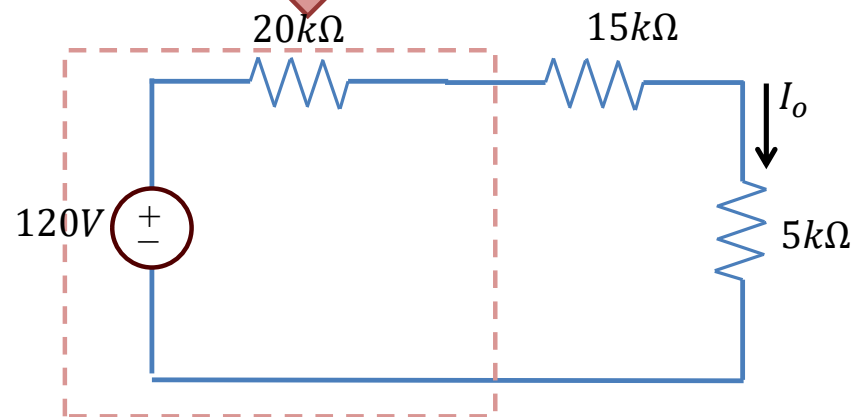
Example 4, cont.



Example 4, cont.



Transform current source

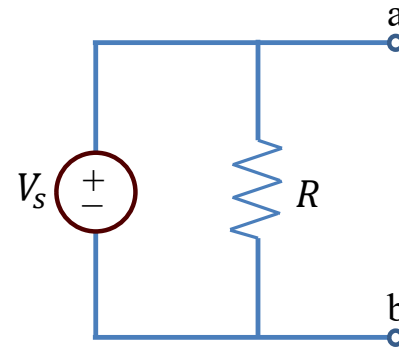
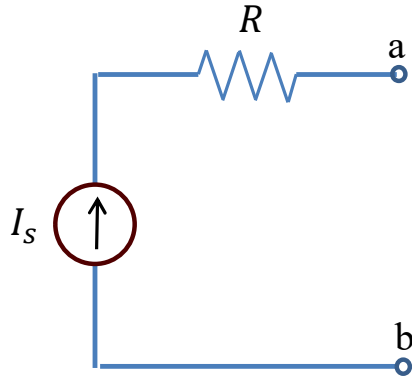


$$I_o = \frac{120V}{40k\Omega} = 3mA$$

Example 4, cont.



- Before we present some examples, it is worth mentioning that we **DO NOT** use source transformations on the following circuits. **WHY NOT?**

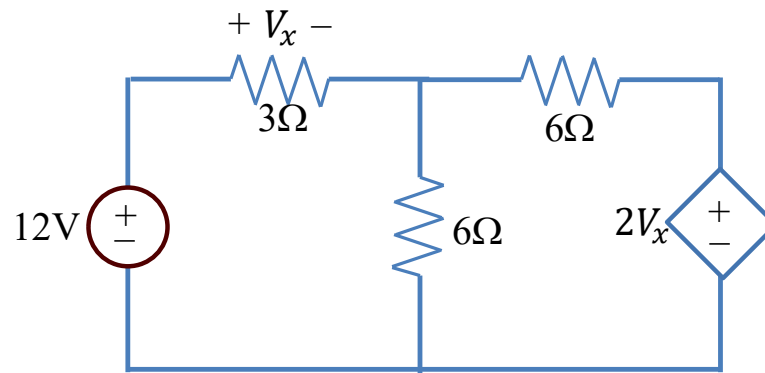


- Q:** What can we do with the above circuits to simplify them? **Hint:** What role does the resistor R play in each circuit above?

Example 5



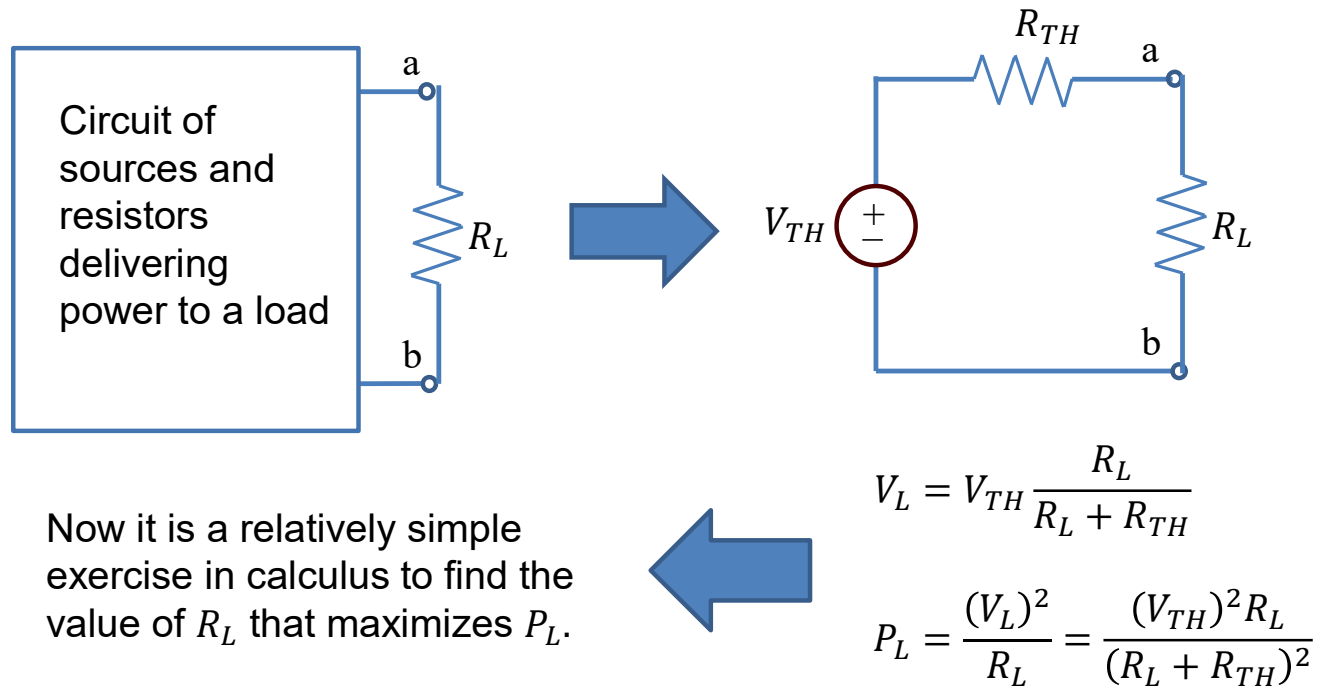
- Find V_x by applying a sequence of source transformations.



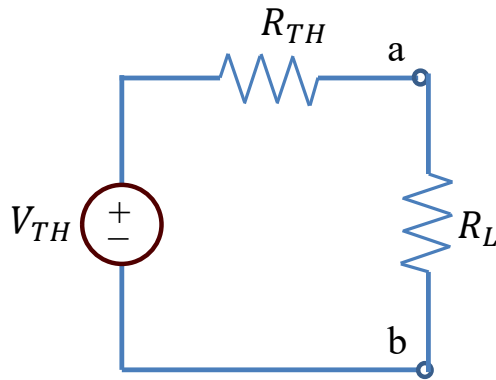
- Note:** If the circuit has dependent sources, we need to make sure to preserve the element with the control voltage/current.

Maximum Power Transfer

- We are often concerned with delivering the most amount of power that we can to a load. We can easily see how to achieve this by using the Thevenin equivalent.



Maximum Power Transfer



$$P_L = \frac{(V_L)^2}{R_L} = \frac{(V_{TH})^2 R_L}{(R_L + R_{TH})^2}$$

$$\frac{dP_L}{dR_L} = (V_{TH})^2 \frac{(R_L + R_{TH})^2 - 2R_L(R_L + R_{TH})}{(R_L + R_{TH})^4} = 0$$

To maximize power delivered to a load, we should choose the load resistance equal to the Thevenin equivalent resistance of the circuit.

$$P_L = \frac{(V_{TH})^2}{4R_{TH}}$$

$$(R_{TH} + R_L)(R_{TH} - R_L) = 0$$

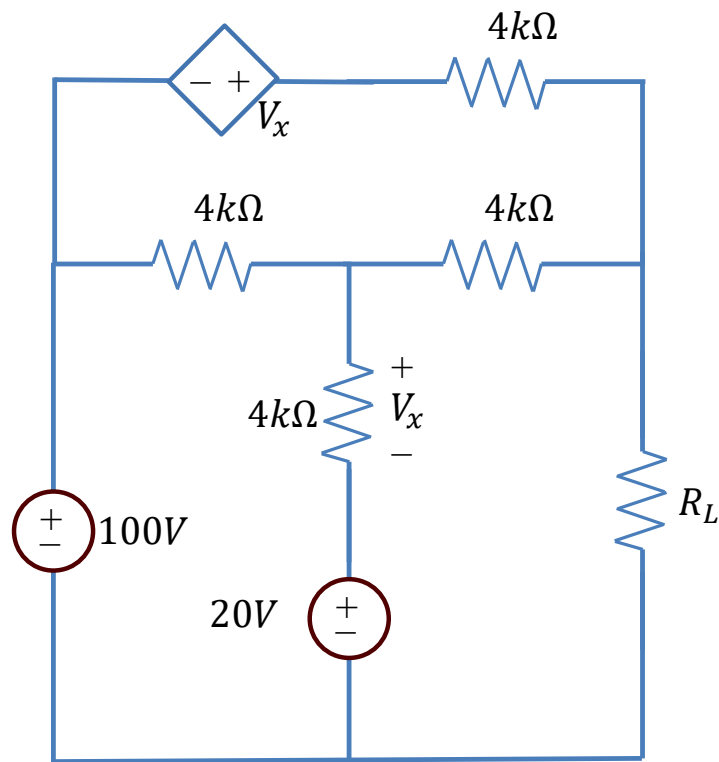
$$R_L = R_{TH}$$



Example 6



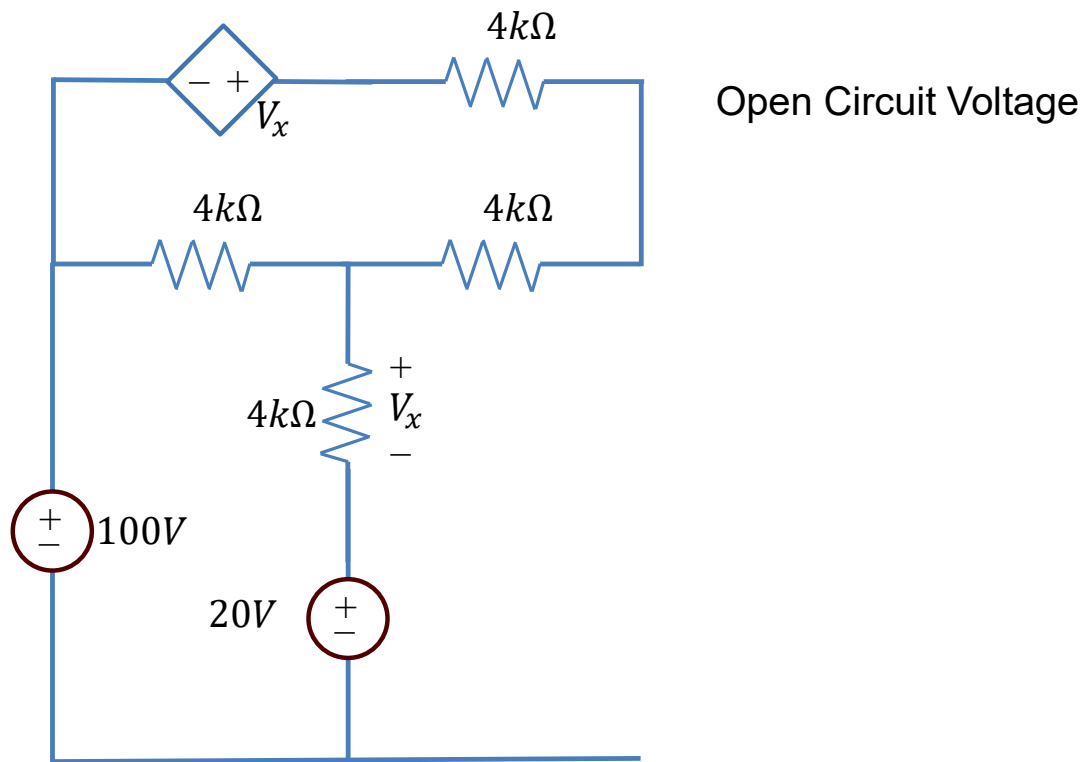
- (a) Find R_L that maximizes power transfer to the load.
(b) What is the maximum power transferred to the load?



Example 6, cont.



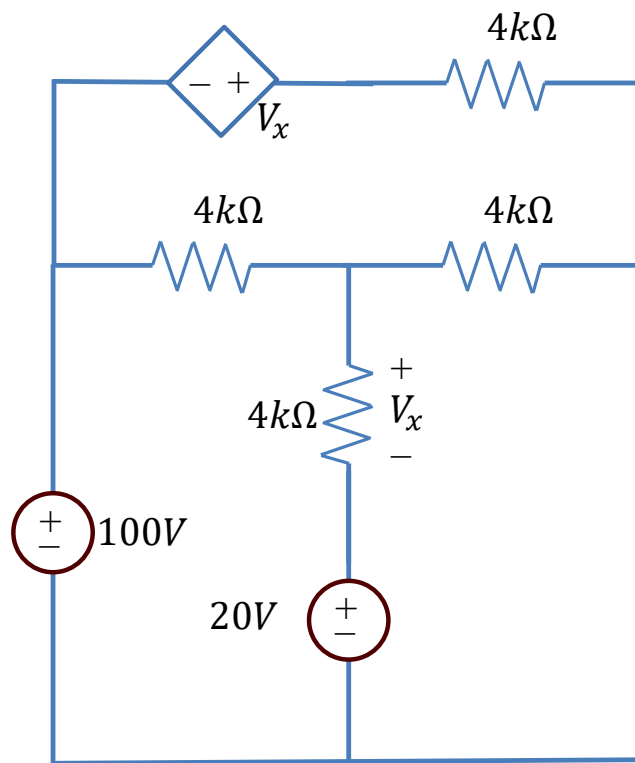
- (a) Find R_L that maximizes power transfer to the load.
- (b) What is the maximum power transferred to the load?



Example 6, cont.



- (a) Find R_L that maximizes power transfer to the load.
 (b) What is the maximum power transferred to the load?

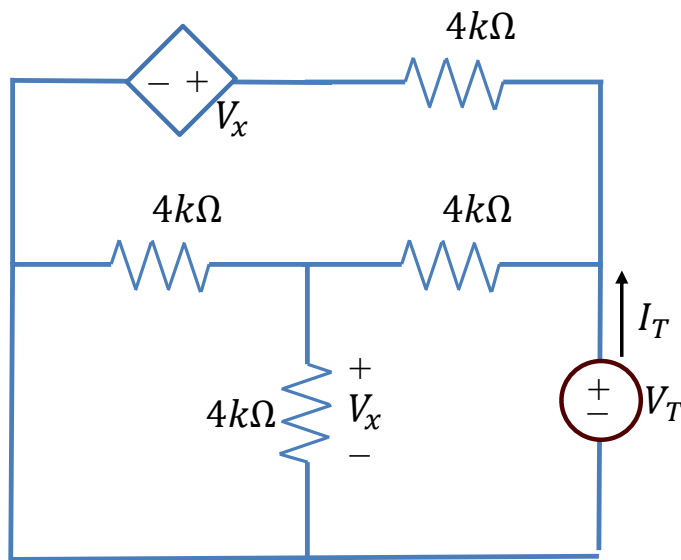


Short Circuit Current

Example 6, cont.



- (a) Find R_L that maximizes power transfer to the load.
- (b) What is the maximum power transferred to the load?



Alternatively (Method 3)
Disable Independent Sources
and Apply Test Source

Next Lecture



- Capacitors and Inductors